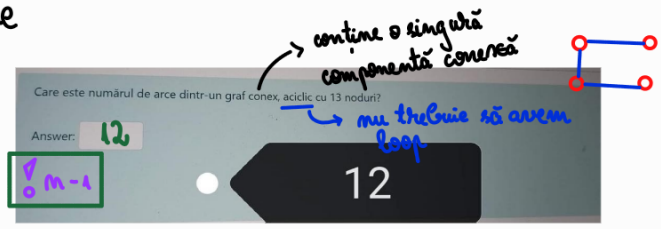
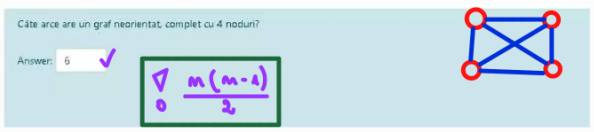
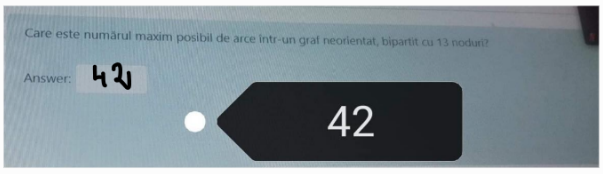
Testie

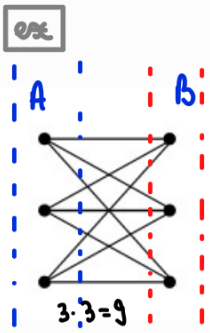
1. 

conține o singură componentă conexă
→ nu trebuie să avem loop
m-1

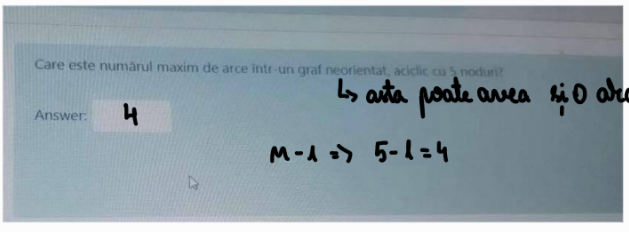
2. 

$\frac{m(m-1)}{2}$

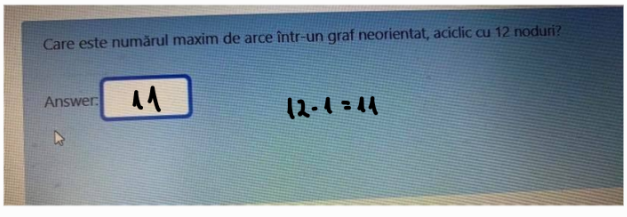
3. 



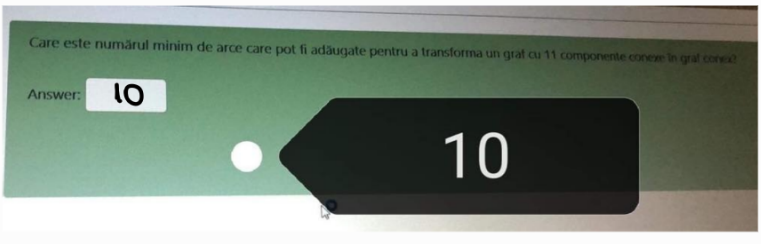
Pres → împărțim să fie cât mai egale
A — 6 noduri
B — 7 noduri
⇒ 6·7 = 42

4. 

↳ asta poate avea și 0 arce
M-1 ⇒ 5-1=4

5. 

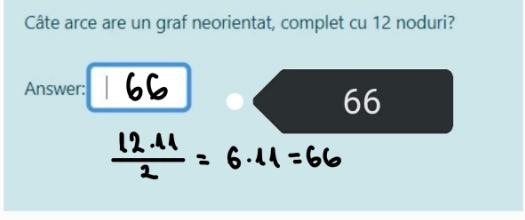
12-1=11

6. 

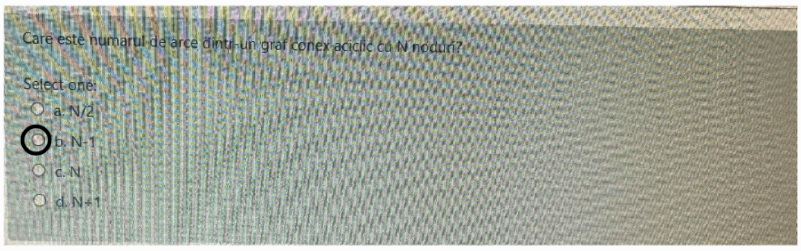
avem 11 noduri singulare (fără niciun arc)
⇒ 11 componente conexe
⇒ min arce pt a deveni un graf conex
= 11-1 = 10

7. 

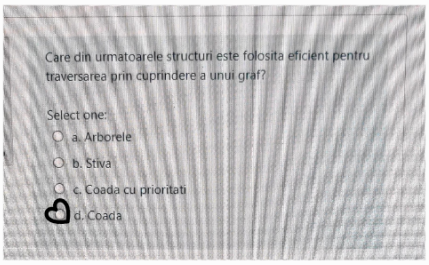
12-1=11

8. 

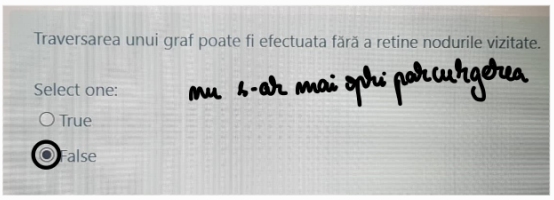
$\frac{12 \cdot 11}{2} = 6 \cdot 11 = 66$

9. 

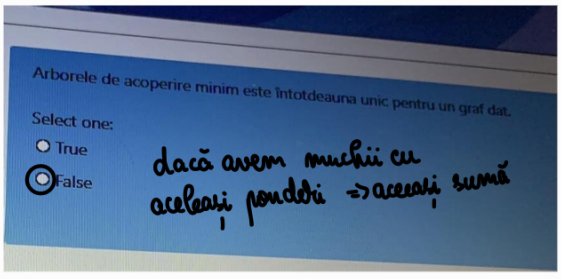
10.



11.



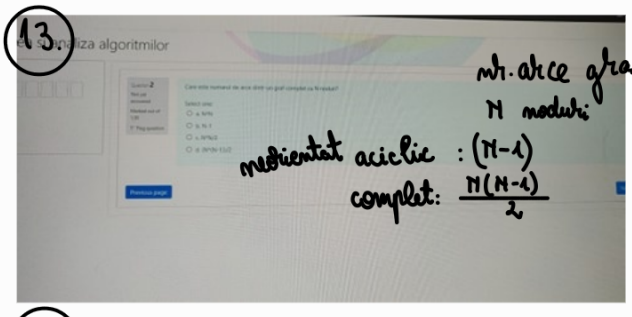
12.



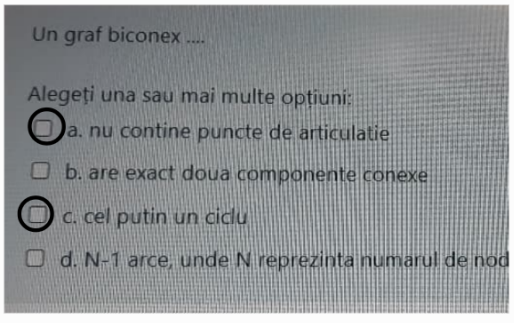
Curs:

Ar putea fi selectat și arcul (5-6) de cost 5. Arborele de acoperire minim **nu** este unic.

13.



14.

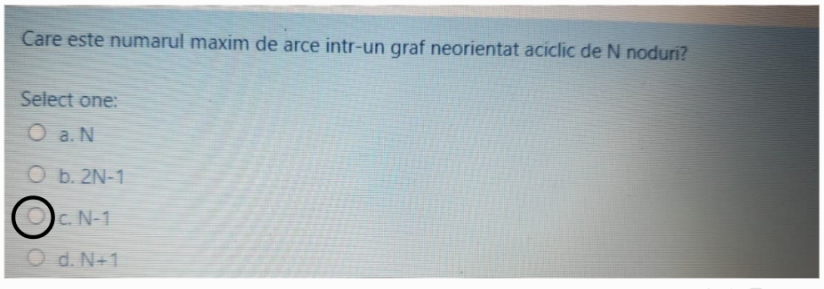


Curs:

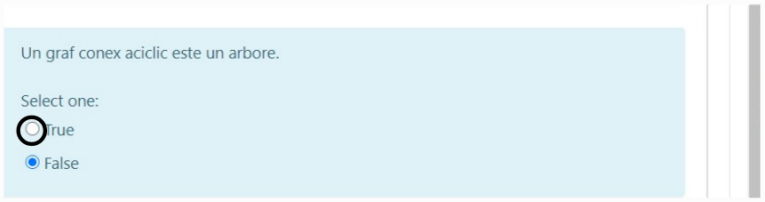
Un graf care **nu** conține **puncte de articulație** se numește **graf biconex**.

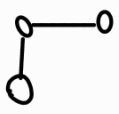
- Într-un **graf biconex**, fiecare pereche de noduri este conectată prin cel puțin **două** drumuri distincte.

15.



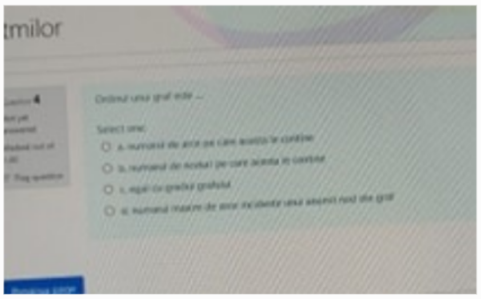
16.





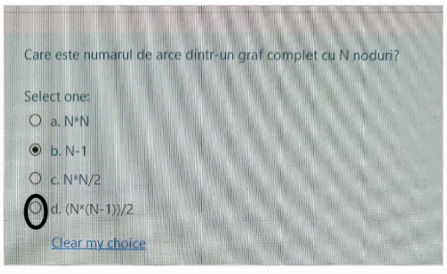
arbore
sau
graf conex aciclic

17.

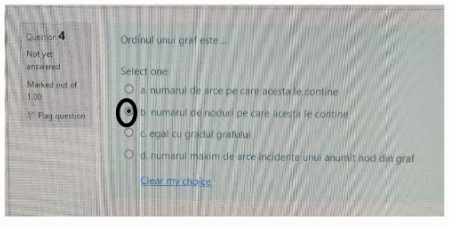


mai jos
Ordinul unui graf

18.

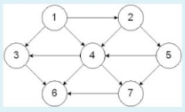


19.



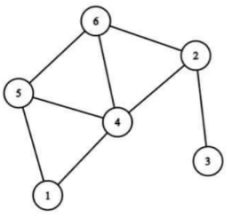
BFS

Consider the graph in the figure below. Write the order that **Depth-First-Search** would explore the nodes **starting from node 1**. Always explore nodes with smaller labels when given the choice. (explore 2 before 3 when you have the choice). Write down: the order in which nodes get visited, and the set of edges which form the DFS-spanning tree.



1 -> 2 -> 4 -> 3 -> 6 -> 7 -> 5

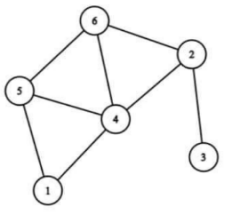
Scrieți nodurile, separate prin virgulă, rezultate după aplicarea unei parcurgeri în adâncime a grafului din imagine, pornind de la nodul 2 (nodurile adiacente vor fi parcurse în ordine crescătoare).



2,3,4,1,5,6

2 -> 3 -> 4 -> 1 -> 5 -> 6

Scrieți nodurile, separate prin virgulă, rezultate după aplicarea unei parcurgeri în adâncime a grafului din imagine, pornind de la nodul 6 (nodurile adiacente vor fi parcurse în ordine crescătoare).

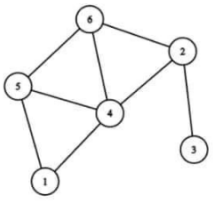


6,2,3,4,1,5

6 -> 2 -> 3 -> 4 -> 1 -> 5

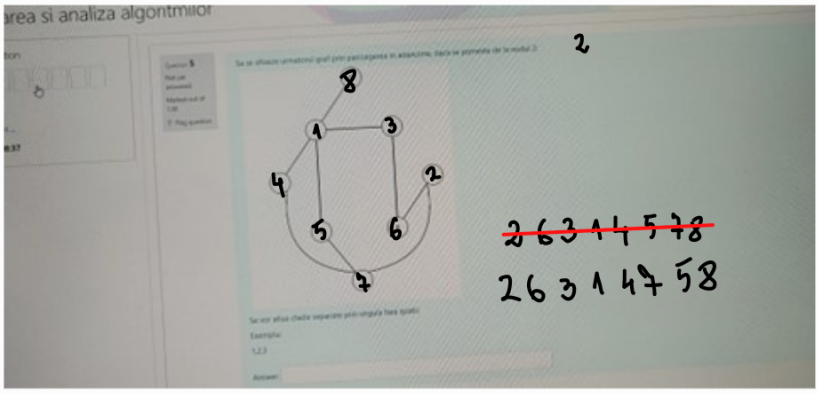
Question 4
Answer saved
Marked out of 2.00
Flag question

Scrieți nodurile, separate prin virgulă, rezultate după aplicarea unei parcurgeri în adâncime a grafului din imagine, pornind de la nodul 2 (nodurile adiacente vor fi parcurse în ordine crescătoare).

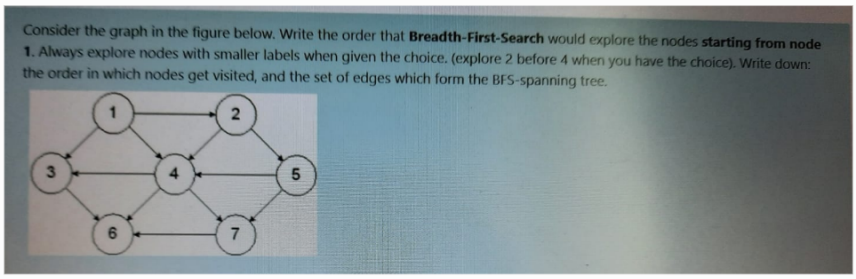


Answer: 2,3,4,1,5,6

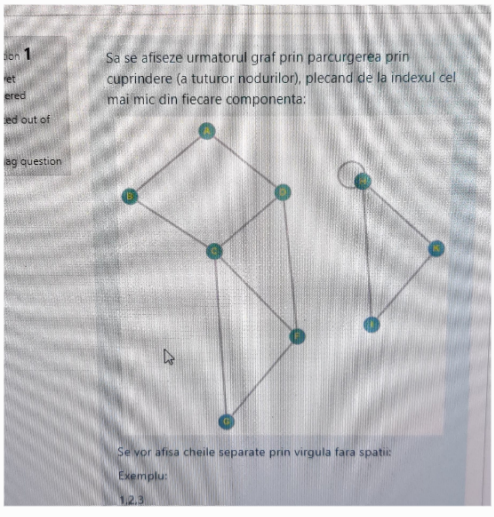
2 -> 3 -> 4 -> 1 -> 5 -> 6


area si analiza algoritmilor
2
8
1
3
4
2
5
6
7
2 6 3 1 4 5 7 8
2 6 3 1 4 7 5 8

BFS



4 5 6 7
3 4 5 6 7
2 3 4 5 6
1 → 2 → 3 → 4 → 5 → 6 → 7



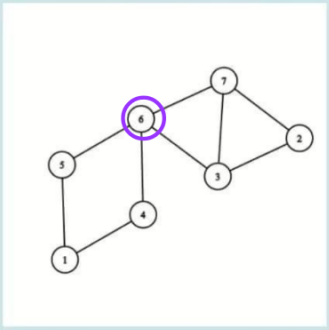
F G
D C F G
B D C F G
A B D C F G H I K

K
I
D C F G K
B D C F I K
A B D C F G H I K

Puncte articulație

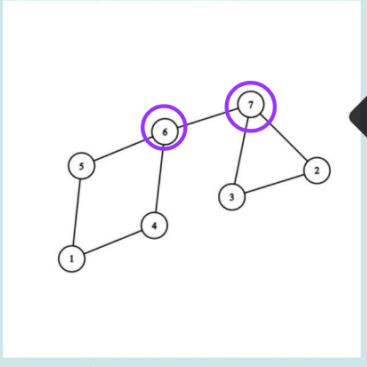
→ nod pe care dacă îl ștergi, graful se împarte în 2 sau mai multe componente conexe

Scrieți nodurile care reprezintă puncte de articulație (ordonate crescător și separate prin virgulă) pentru graful din imagine.



Answer: 6 ✓

Scrieți nodurile care reprezintă puncte de articulație (ordonate crescător și separate prin virgulă) pentru graful din imagine.



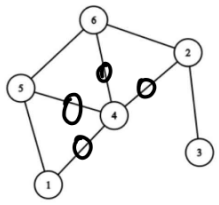
6,7 ✓

Answer: 6,7

Grad graf

∇° = cel mai mare grad al unui nod
= nr. maxim de arce care pleacă dintr-un nod

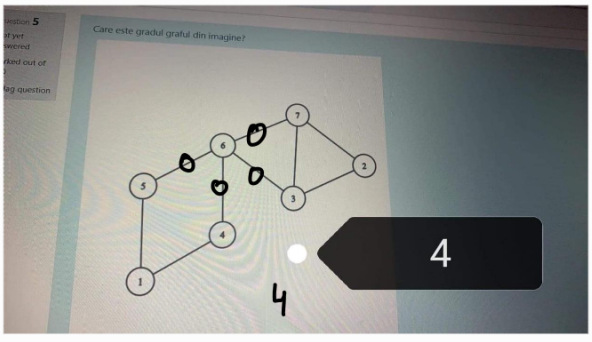
Care este gradul grafului din imagine?



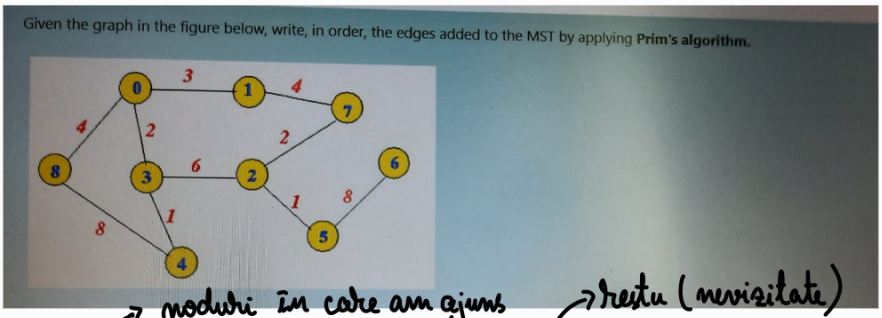
4

4

Answer:



Prim's alg



V → noduri în care am ajuns | N-V → restu (nevizitate) | arc → luăm cel mai mic cost

V	N-V	arc
{0}	{1,2,3,4,5,6,7,8}	0-3
{0,3}	{1,2,4,5,6,7,8}	3-4
{0,3,4}	{1,2,5,6,7,8}	0-1
{0,1,3,4}	{2,5,6,7,8}	0-8 (crescător ales 0)
{0,1,3,4,8}	{2,5,6,7}	1-7
{0,1,3,4,7,8}	{2,5,6}	7-2
{0,1,2,3,4,7,8}	{5,6}	2-5
{0,1,2,3,4,5,7,8}	{6}	5-6
{0,1,2,3,4,5,6,7,8}		

Order ↓

Kruskal's algorithm

noduri	arc
{0}{1}{2}{3}{4}{5}{6}{7}{8}	2-5 (luăm ponderea cea mai mică)
{0}{1}{2,5}{3}{4}{6}{7}{8}	3-4
{0}{1}{2,5}{3,4}{6}{7}{8}	0-3
{0,3,4}{1}{2,5}{6}{7}{8}	2-7
{0,3,4}{1}{2,5,7}{6}{8}	0-1
{0,1,3,4}{2,5,7}{6}{8}	0-8
{0,1,3,4,8}{2,5,7}{6}	1-7
{0,1,2,3,4,5,7,8}{6}	5-6 (este ultimul neagrupat → nodul 6)
{0,1,2,3,4,5,6,7,8}	

Order ↓

Lungime drum minim



! graf orientat

A -> B

A, B, E, B

A, b, E

A, b, G, i, H

	A	B	C	b	E	F	G	H	i
b	0	~~10~~ 8+1	~~30~~ 5+5	8+1	5	5+3	5+5	5+1	7+1	~~5+3~~ 6+1
T	A	~~A~~ E	~~A~~ b	H	A	b	b	b	i	~~b~~ G

=> A -> b -> E -> B cost: 5 + 3 + 1 = 9

	A	B	C	b	E	F	G	H	i
b	0	~~10~~ 8+1	~~30~~ 5+5	8+1	5	~~∞~~ 5+3	~~∞~~ 5+5	~~∞~~ 5+1	~~∞~~ 7+1	~~∞~~ ~~5+3~~ 6+1
T	A	~~A~~ E	~~A~~ b	A	b	b	b	i	~~b~~ G

b - C -> 5 + 5 = 10 < 30 actualizăm

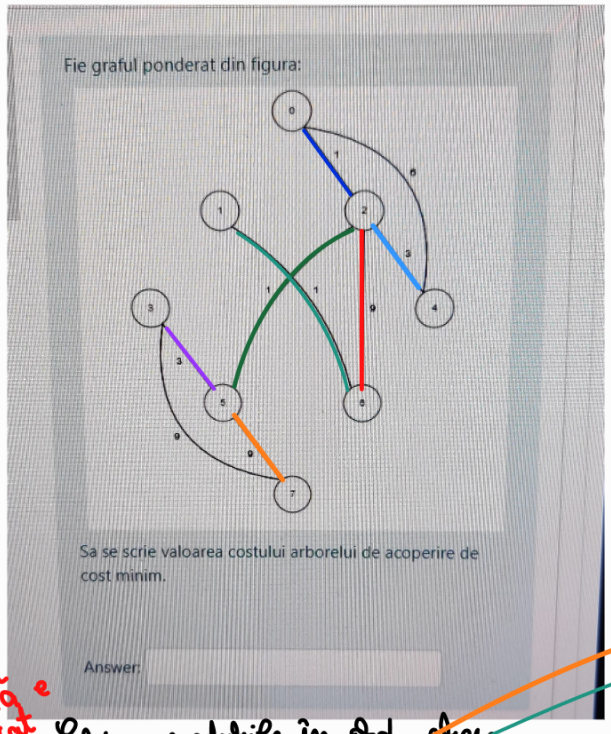
E - b -> 8 + 1 = 9 < 10 actualizăm

H - C -> 8 + 1 = 9 < 10 actualizăm

lungime drum A -> B

A b E B cost: 5 + 3 + 1 = 9

Cost



Dacă e orientat (asta nu e)

Luăm pe rând nodurile în ord. crescătoare.

0, 2, 5, 7

0, 2, 6

∞ - nu putem ajunge

luăm mereu cost minim

	0	1	2	3	4	5	6	7
	suntem în el							
d (cost)	0	~~∞~~ 10+1	1	~~∞~~ 2+3	~~6~~ 3+1	~~∞~~ 1+1	~~∞~~ 1+9	~~∞~~ 2+9
T (ultim nod înainte de ce e sus)	0	6	0	5	~~2~~ 2	2	2	5

dacă costul e mai mare, nu actualizăm

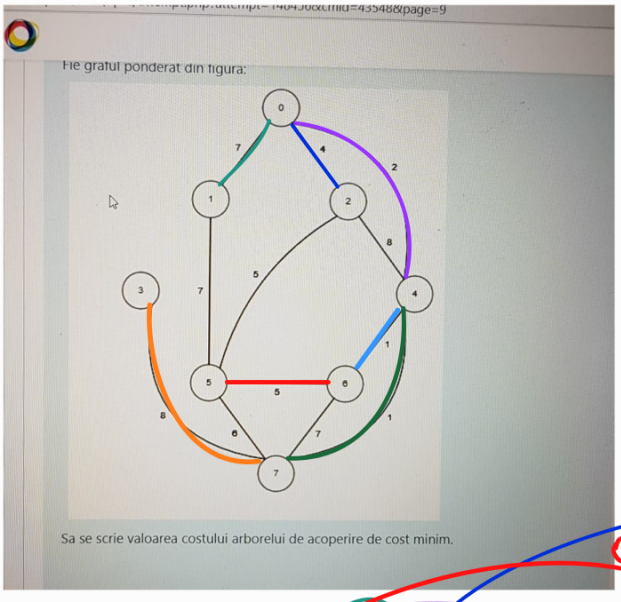
7 - 3 => 11 + 9 = 20 > 5

0 - 4 e parcurs mai devreme

cost: 1 + 3 + 9 + 1 + 1 + 9 + 3 = 4 + 20 + 3 = 27

Dacă e neorientat → Prim sau Kruskal

U	N - U	arce	cost
{0}	{1,2,3,4,5,6,7}	0-2	1
{0,2}	{1,3,4,5,6,7}	2-5	1
{0,2,5}	{1,3,4,6,7}	3-5	3
{0,2,3,5}	{1,4,6,7}	2-4	3
{0,2,3,4,5}	{1,6,7}	2-6	9
{0,2,3,4,5,6}	{1,7}	6-1	1
{0,1,2,3,4,5,6}	{7}	3-7	9
{0,1,2,3,4,5,6,7}			27



luăm în ord. crescătoare

0 4,6,5

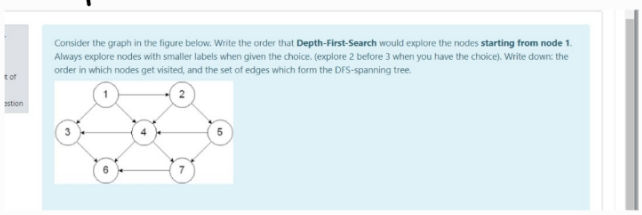
	0	1	2	3	4	5	6	7
b	0	7	4	~~∞~~ 3+8	2	~~∞~~ 3+5	~~∞~~ 2+1	~~∞~~ 2+1
T	0	0	0	7	0	6	4	4

2-4 → 2+8=10>4 nu actualizăm
6-7 → 3+7=10>3 nu actualizăm
4-5 → 3+6=9>8 nu actualizăm
2-5 → 4+5=9>8 nu actualizăm
1-5 → 7+7=14>8

cost: 7+4+2+1+1+5+8 = 11+4+13 = 28

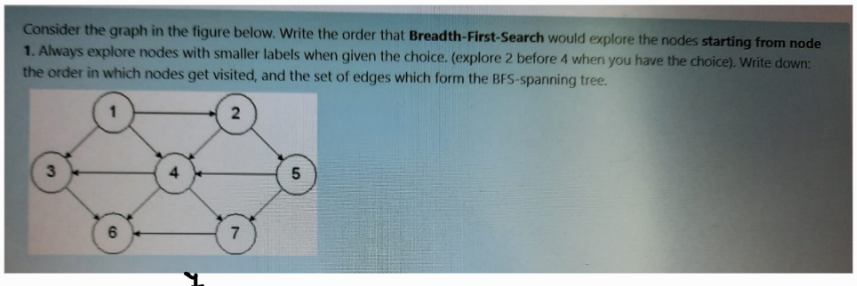
U	N-U	arc	cost
{0}	{1,2,3,4,5,6,7}	0-4	2
{0,4}	{1,2,3,5,6,7}	4-6	1
{0,4,6}	{1,2,3,5,7}	4-7	1
{0,4,6,7}	{1,2,3,5}	0-2	4
{0,2,4,6,7}	{1,3,5}	2-5	5
{0,2,4,5,6,7}	{1,3}	0-1	7
{0,1,2,4,5,6,7}	{3}	7-3	8
{0,1,2,3,4,5,6,7}			28

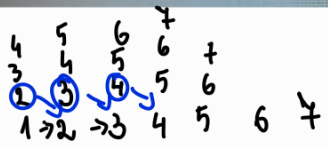
Recapitulare

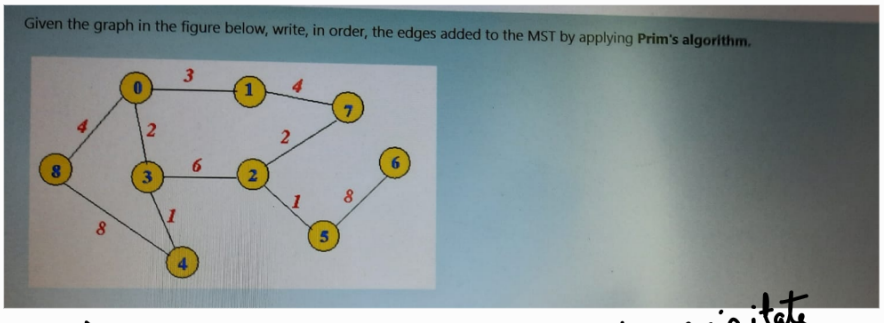


1 → 2 → 4 → 3 → 6 → 7 → 5

BFS







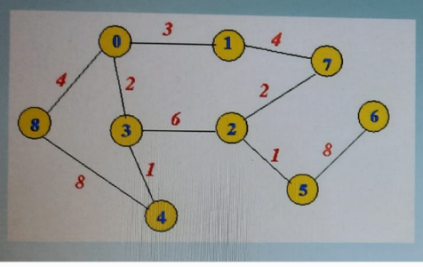
Prim's algorithm

U → noduri vizitate

V - U → noduri nevizitate

U	V - U	arc ales
{0}	{1,2,3,4,5,6,7,8}	0 - 3
{0,3}	{1,2,4,5,6,7,8}	3 - 4
{0,3,4}	{1,2,5,6,7,8}	0 - 1
{0,1,3,4}	{2,5,6,7,8}	0 - 8
{0,1,3,4,8}	{2,5,6,7}	1 - 7
{0,1,3,4,7,8}	{2,5,6}	7 - 2
{0,1,2,3,4,7,8}	{5,6}	2 - 5
{0,1,2,3,4,5,7,8}	{6}	5 - 6
{0,1,2,3,4,5,6,7,8}	{}	

Given the graph in the figure below, write, in order, the edges added to the MST by applying **Prim's algorithm.**



Kruskal's alg.

{0}{1}{2}{3}{4}{5}{6}{7}{8}

{0}{1}{2,5}{3}{4}{5}{6}{7}{8}

{0}{1}{2,5}{3,4}{5}{6}{7}{8}

{0,3,4}{1}{2,5}{6}{7}{8}

{0,1,3,4}{2,5}{6}{7}{8}

{0,1,3,4,8}{2,5}{6}{7}

{0,1,3,4,7,8}{2,5}{6}

{0,1,2,3,4,5,7,8}{6}

{0,1,2,3,4,5,6,7,8}

arc.

2-5

3-4

0-3

2-7

0-8

1-7

3-2

5-6